



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

SPECTROSCOPY OF ASTROPHYSICAL PLASMAS

Fritz Ali Agildere

June 9, 2026



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.



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Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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- Dynamically Evolving Obscuring Starforming Structure

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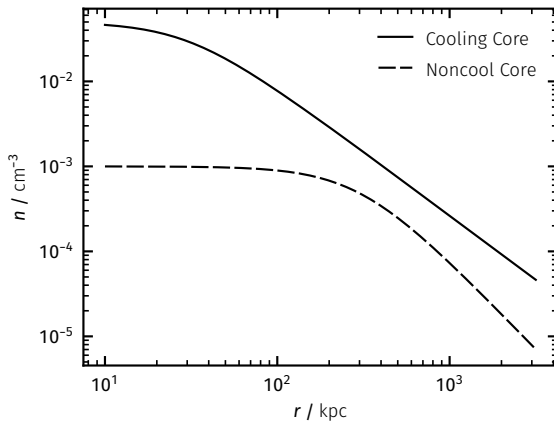
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- Embedded Stellar Remnant Thermal Effects

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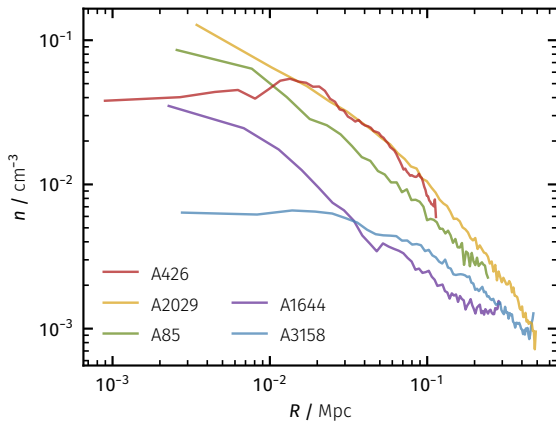
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- Statistically Consistent Viewing Angle Sampled Distributions

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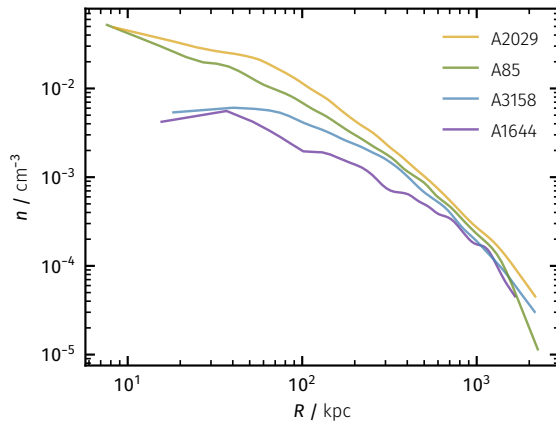
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- CO SLED Molecular Gas Tracers (ALMA)
- IR Starforming & Disk Signatures (JWST)

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)